

**Project Title:** AR-MULTI++ Web:Context-Aware Multilingual Scene Text Recognition, Translation, and Voice Output with Augmented Reality Simulation.

## **System Requirements**

The AR-MULTI++ Web system requires both hardware and software components to ensure smooth execution of the multilingual text recognition, translation, and speech generation processes. The requirements are categorized as follows:

### **1. Hardware Requirements**

The system does not require high-end hardware, but a standard computing device with the following specifications is sufficient for efficient performance:

- **Processor:** Intel Core i3 or above
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** At least 10 GB free disk space
- **Display:** Standard monitor with internet browser support
- **Input Devices:** Keyboard and mouse
- **Internet Connectivity:** Required for cloud-based APIs (OCR, Translation, TTS)

### **2. Software Requirements**

The system is developed using modern web technologies and requires the following software components:

- **Operating System:** Windows 10/11, Linux, or macOS
- **Frontend:** Next.js, React.js
- **Backend:** Node.js, Express.js
- **Programming Languages:** JavaScript
- **Database/Storage:** Local storage or file system (for uploads)
- **Browser:** Google Chrome, Microsoft Edge, or any modern web browser

### **3. Tools and Technologies Used**

The system integrates multiple tools and APIs for processing:

- **OCR Tool:** Google Cloud Vision API (for text extraction)
- **Translation API:** Google Translate API
- **Text-to-Speech:** Google Text-to-Speech API
- **Development Tools:** Visual Studio Code
- **Version Control:** Git (optional)

#### 4. Functional Requirements

The system must be capable of performing the following functions:

- Accept image input from the user
- Extract text from images using OCR
- Detect the source language automatically
- Translate extracted text into a target language
- Generate speech output for translated text
- Create overlay images with translated text
- Display results including text, image, and audio

#### 5. Non-Functional Requirements

The system should satisfy the following conditions:

- **Performance:** Fast processing and real-time response
- **Accuracy:** High accuracy in OCR and translation
- **Usability:** Simple and user-friendly interface
- **Scalability:** Ability to handle multiple users and requests
- **Reliability:** Stable performance with minimal errors